PHYSICAL CUME
OCTOBER 29, 2004

The well-known van der Waals equation of state connects the absolute temperature T, the molar volume V, and the pressure P of a real gas by the equation

$(P + a/V^2)(V - b) = RT$

where a and b are constants for a given gas. For CO_2, a=3.6068 atm L^2 and b=0.0428 L; the absolute zero is 0 K = -273.2 C^o.

The gas constant, R=0.082057 L atm mol^{-1} K^{-1}. This equation is valid if there is only a single phase, not a liquid and a gas together.

1) What values of the constants *a* and *b* make the van der Waals equation identical with the equation of state of the ideal (perfect) gas?

2) Under what conditions (temperature and pressure) does the van der Waals equation approximate most closely the equation of state of the ideal gas (choose one):

(a) high temperature and high pressure
(b) high temperature and low pressure
(c) low temperature and high pressure
(d) low temperature and low pressure

(3) The pressure of CO_2 (in atm) as a function of the molar volume (in L) is given in the graph below, together with the values predicted by the van der Waals equation, at three different temperatures: 46.8 C^o, 26.8 C^o and 6.8 C^o. The van der Waals equation values are the continuous curves, the real values are the dots. The dashed line separates the region where the gas and liquid coexist from the region where there is only one phase.
(a) Explain the why the pressure of the real system is constant at a given temperature within the two-phase region
(b) Based on this diagram, what is the critical temperature and critical pressure of CO_2?
(c) Define the critical point of a gas

(4) The pressure of a real physical system always diminishes if the volume is increased. However, as shown in the graph, the van der Waals equation can show an unphysical increase in the pressure if the volume decreases. Derive the condition for this to happen (Hint: express the fact that P increases if V increases mathematically. This will automatically give the condition.)